

Darshan Poudel

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EDUCATION

University of Regina

B.Sc. Computer Science

Regina, Canada

2030

- **Publications:** [ARIA: Adaptive Recurrent Intelligence Architecture](#) (Continual Learning), [NCG: Novelty-triggered Capacity Growth](#).
- **Awards & Honors:** 3× Hackathon Winner ([Virtu Hack](#), [Code Rush](#), [Student Hacks](#)), [Ace Ardents 2024 Winner](#).

EXPERIENCE

Google

Open Source Contribution | [Syzkaller](#)

- Fixed musl build incompatibility by adding pkey syscall compatibility shims. Implemented platform-specific solutions for pkey_alloc, pkey_free, and pkey_mprotect across musl/glibc and amd64/arm64 architectures, enabling executor builds on Alpine Linux and other musl-based systems.

Anthropic

Open Source Contribution | [Buffa](#)

- Designed and implemented buffa-remote-derive macro crate enabling foreign types as custom field types. 1,317-line feature spanning 22 files with sophisticated two-tier trait design (ProtoString, ProtoBytes, ProtoList, ProtoBox, MapStorage) and comprehensive integration tests.

NVIDIA

Open Source Contribution | [infra-controller](#)

- Implemented intelligent health report suppression with time-based caching and content hashing. Per-key deduplication skips unchanged success-only reports while preserving alert forwarding, reducing unnecessary API traffic for stable machines while maintaining system responsiveness.

Zeno Enterprises

Software Engineer

Pokhara, Nepal

July 2025 – May 2026

- Architected a 7-module ML pipeline that transformed raw biometric sensor data into predictive health intelligence, combining sleep/stress models, anomaly detection (Isolation Forest), LSTM-based forecasting, personalized baselines, and LLM-powered health coaching via OpenRouter API.
- Built production FastAPI backend serving real-time health analytics to iOS/Android clients with TimescaleDB for time-series optimization, enabling efficient querying of billions of biometric data points.
- Implemented privacy-focused architecture with secrets management, TLS encryption, and per-user data isolation; added structured logging and alerting for 40% faster incident detection.

FEATURED PROJECTS

ARIA

Continual Learning Architecture

June 2026

- Achieves 98.50% accuracy with 0.86% forgetting on Split-MNIST, outperforming EWC (97.35%, 2.29%) by 62% less forgetting and 1.15 points accuracy. Combines morphogenic attention, plasticity-gated MLPs, architecture genome vectors, cognitive budget allocation, and slow-pathway consolidation for dynamic capacity allocation.

DuckAI

AI Contextual Assistant

Sep 2025

- Cross-platform desktop assistant providing real-time contextual help by analyzing screen and work context via OCR. Built with Electron (desktop), React (UI), and FastAPI (backend). Enables always-on support without manual input or app switching.

FieldSense

Agricultural ML Decision Platform

August 2024

- Intelligent farming platform with ML-driven crop recommendations, fertilizer optimization algorithms, and computer vision leaf disease detection at 95% accuracy. Production REST API with explainable reasoning, regulatory audit trails, and persistent history. Deployed with live demonstrations.

TECHNICAL SKILLS

Languages & Frameworks: Go, Python, Java, Rust, JavaScript, SQL, Flask, FastAPI, Node.js, React, JUnit

ML, Databases & Infra: PyTorch, HuggingFace Transformers, TRL, NumPy, Continual Learning, PostgreSQL, Redis, Docker, Kubernetes, ArgoCD, GCP